



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

Our approach encompassed API integration and webscraping for data collection. Next, preprocessing data were using python. SQL were used to extract information from the dataset. Data visualization and trend analysis were generated to give a quick insights. Finally, machine learning models were used to predict the successful landing.

- Summary of all results

we analyzed the data from SpaceX launch and find the correlations among variables which influence the success of landing. Thus, we developed a predictive model to forecast the landing. The model achieved an impressive accuracy of 90% in this task.

Introduction

- SpaceX advertises Falcon 9 rocket launches is cheaper than others and they reuse it to make sure the cost is saving. An analysis of successful rate for this retrieval events is valuable for evaluation efficiency. The analysis can offer insights in decision-making phase to enhance efficiency.
- Our objective is to predict the success landing of SpaceX launch. This contributes to save resources.

Section 1

Methodology

Methodology

Executive Summary

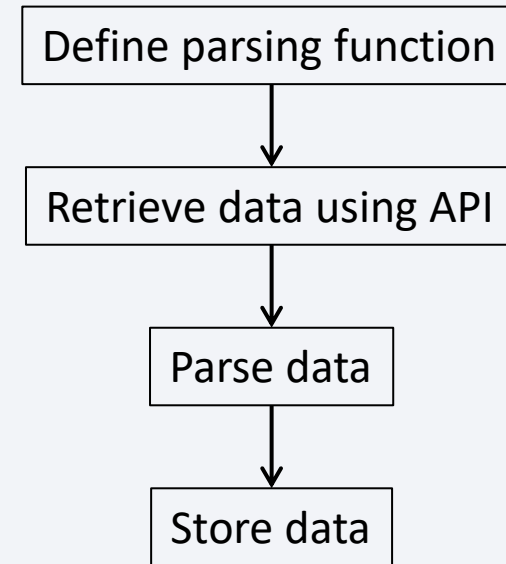
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- The data collection will be discussed in the following slides

Data Collection – SpaceX API

1. Define auxiliary function to parse the data
2. Retrieve data from the **REST API** using **GET**
3. Parse the data with the previously built auxiliary function
4. Store the data in pandas dataframe
5. The result was saved as **CSV**



GitHub URL of the completed SpaceX API:

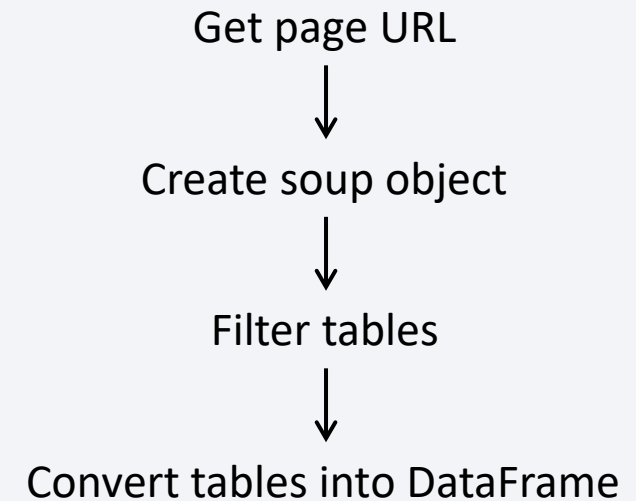
(SpaceX data collection API)

Data Collection - Scraping

1. Request data with **request.get** method to download page code
2. Create a **BeautifulSoup** object to manipulate the html text
3. Filtered the desired table with 'table' then iterated through the table with 'th'
4. Converted the data from **HTML** into a **Pandas DataFrame**
5. The result was saved as **CSV**

GitHub URL of the completed web scraping:

(SpaceX data webscraping)

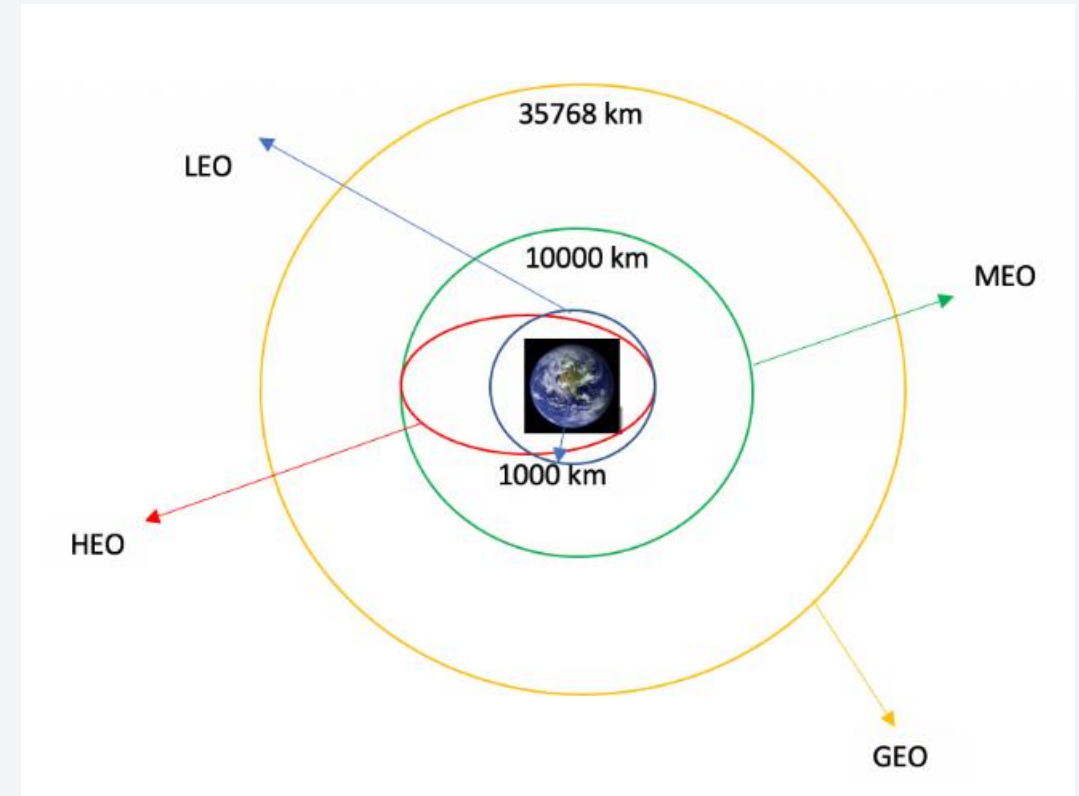


Data Wrangling

- The number of launches on each sites was calculated
- The number of occurrence of each orbit was calculated
- The number and occurrence of mission outcome of the orbits was calculated
- The landing outcome was created
- The result was saved as **CSV**

GitHub URL of the completed web scraping:

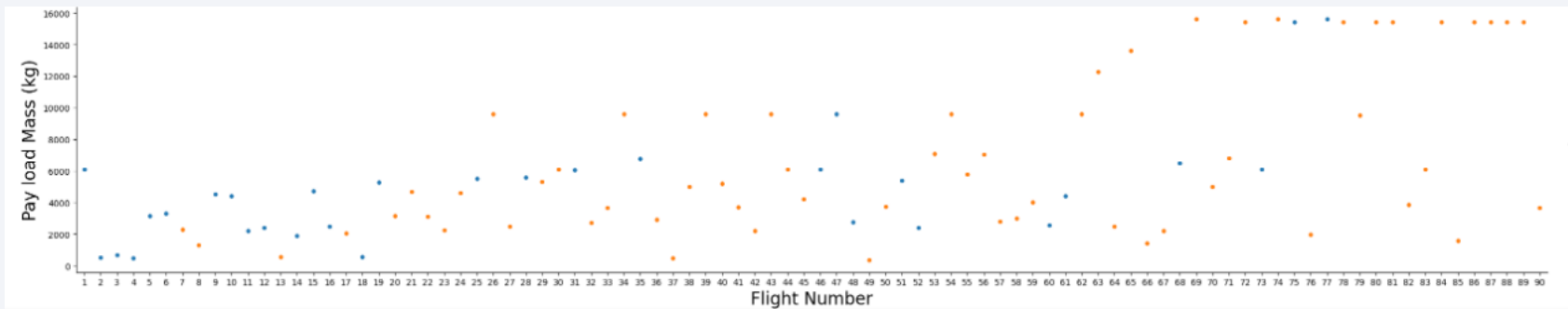
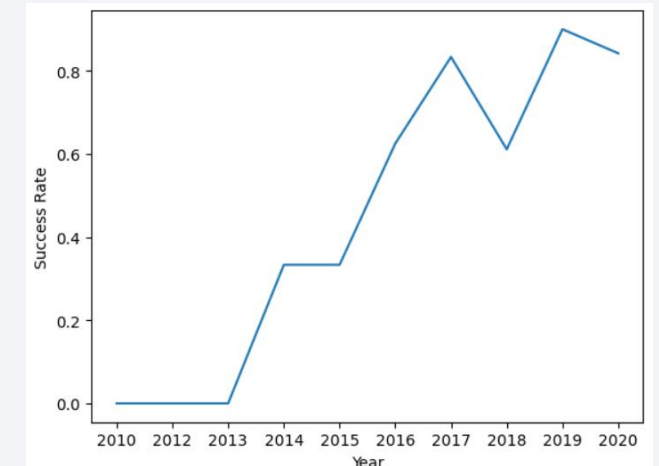
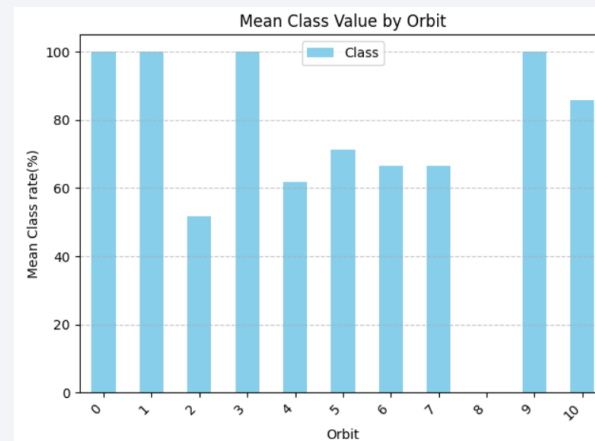
(SpaceX data wrangling)



EDA with Data Visualization

- The data were visualized using several features (ex: pay load mass, flight number, class , orbit, success rate, year)
- Scatter plot, bar plot, line plot were created

GitHub URL of the completed Visualization:
(EDA with Visualization)



EDA with SQL

- Find the names of launch sites
- Find launch sites begin with the string 'CCA'
- Find the total payload mass carried by boosters launched by NASA (CRS)
- Find average payload mass carried by booster version F9 v1.1
- Find the date the first successful landing outcome (ground pad)
- Find the names of the boosters success (drone ship) and $4000 < \text{payload mass} < 6000$
- Find the total number of successful and failure mission outcome
- Find the booster_versions with the maximum payload mass
- Find the month names, failure landing_outcomes (drone ship) ,booster versions, launch_site for the months in year 2015
- Find the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order

Build an Interactive Map with Folium

- All launch sites were marked and map objects were added
- The feature launch outcomes were assigned (failure=0, success=1)
- The colored labeled marker cluster shows successful rate
- The distance between launch sites were calculated for answering questions
 - launch sites close to railways, highways, coastlines
 - launch sites are away from cities

GitHub URL of the completed Interactive Visual Analytics with Folium : (launch-site-location Folium)

Build a Dashboard with Plotly Dash

- An interactive dashboard with plotly dash was created
- Pie charts show the launches in certain sites
- Scatter graph shows the relationship with outcome and payload mass in different booster version

GitHub URL of the completed an Interactive Dashboard with Plotly Dash:
([launch-site-location Folium](#))

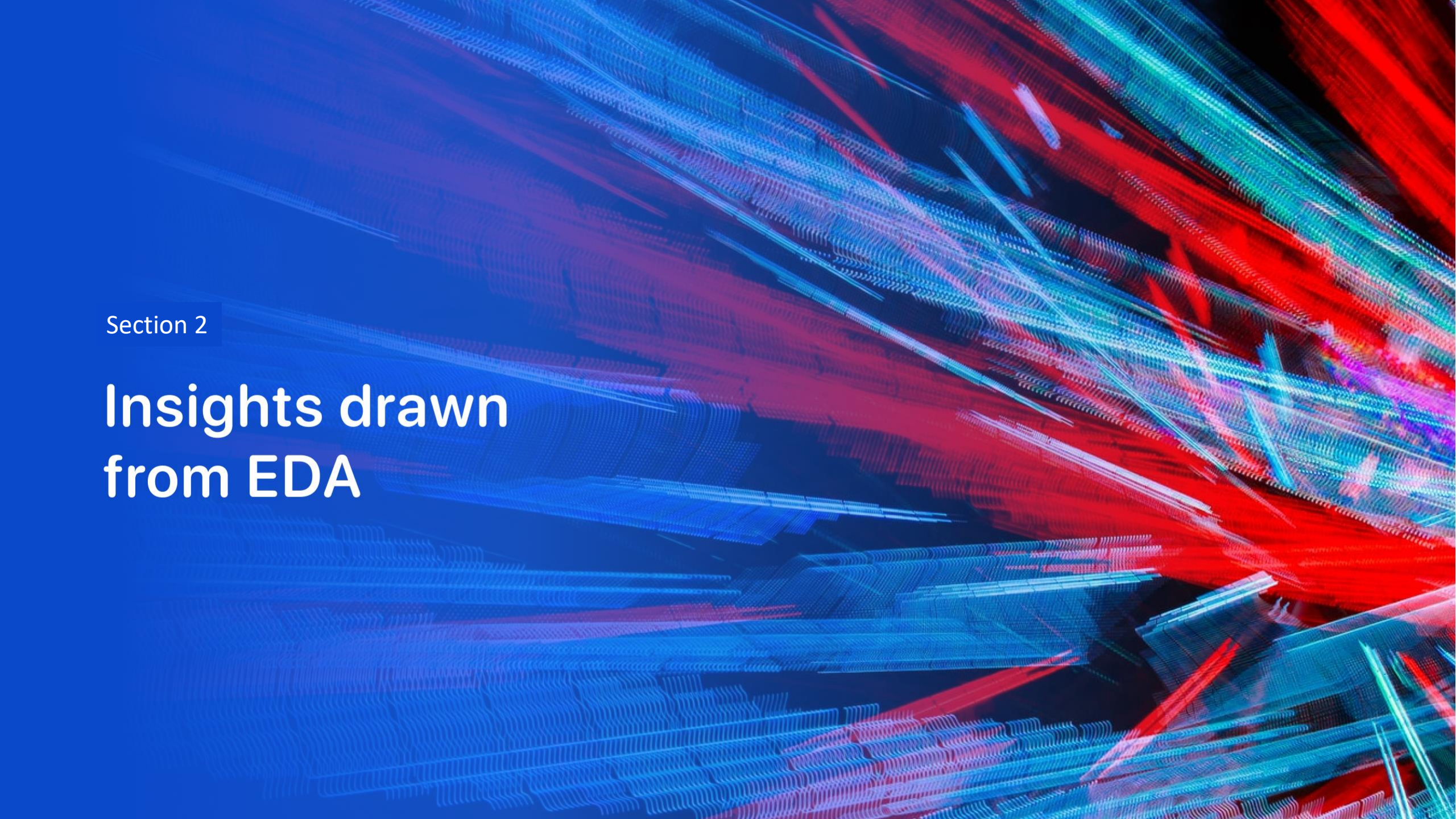
Predictive Analysis (Classification)

- The data were loaded using numpy and pandas to transform, split into training and testing data
- Different machine learning models and tune different hyperparameters using **GridSearchCV** were used
- Accuracy was the metric of our model

GitHub URL of the completed the Machine Learning Prediction:
(SpaceX Machine Learning Prediction)

Results

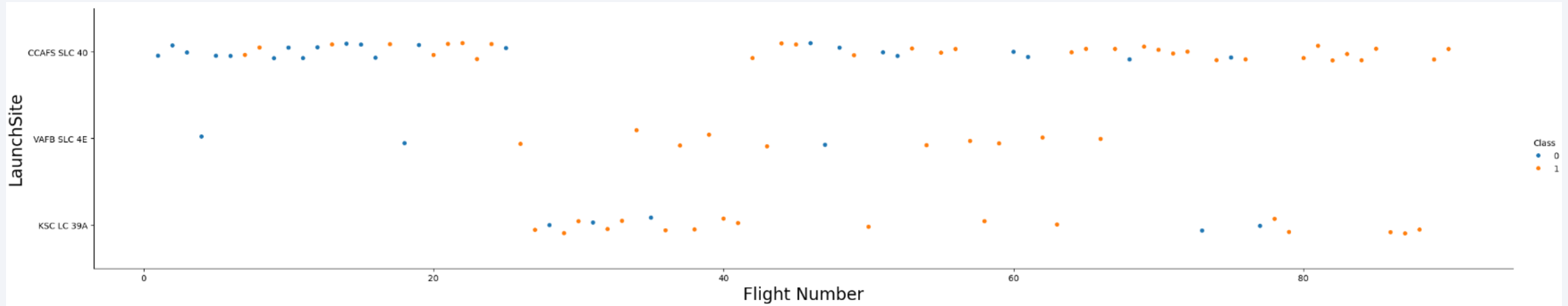
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

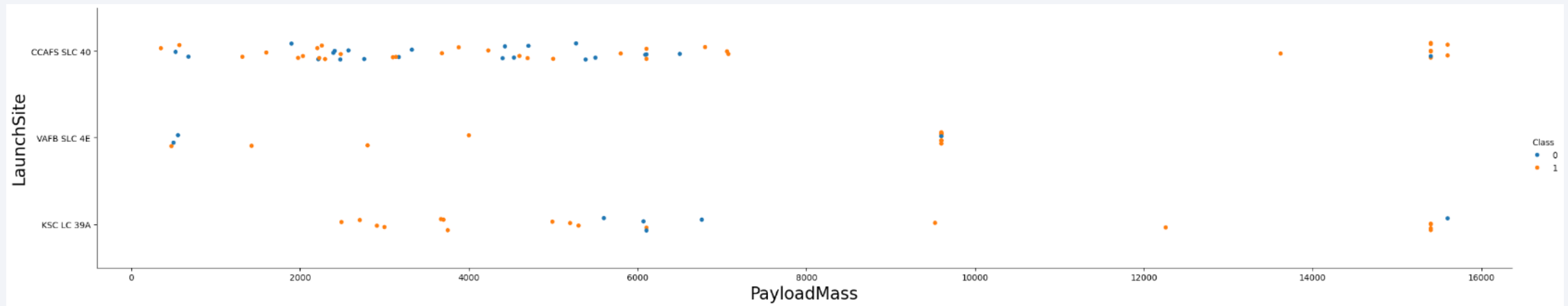
Insights drawn from EDA

Flight Number vs. Launch Site



- This plot shows launch sites (CCSFS SLC 40, KSC LC 39A, and VAFB SLC 4E) with flight number, and class(Failure, Success)
- CCSFS SLC 40 has more launch activity

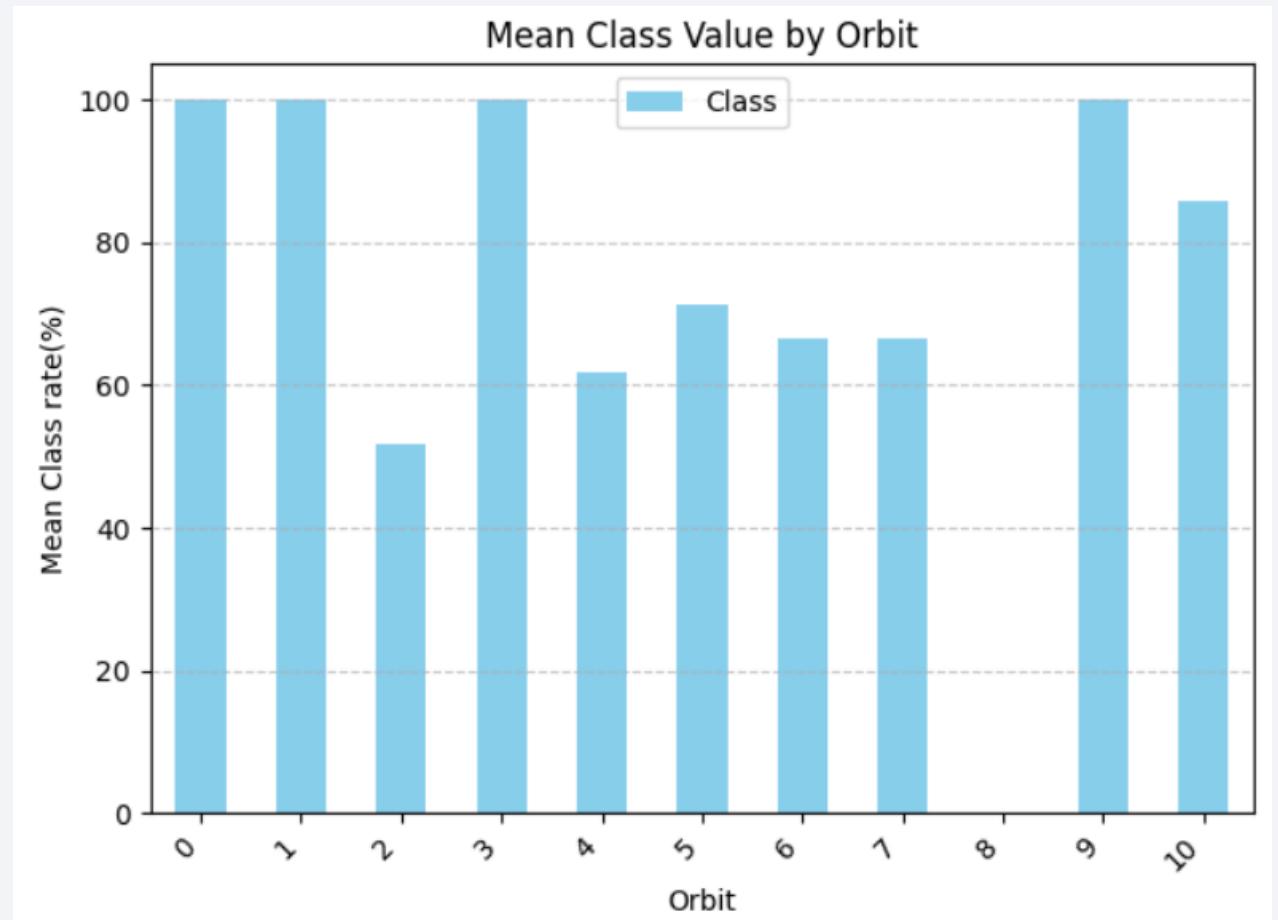
Payload vs. Launch Site



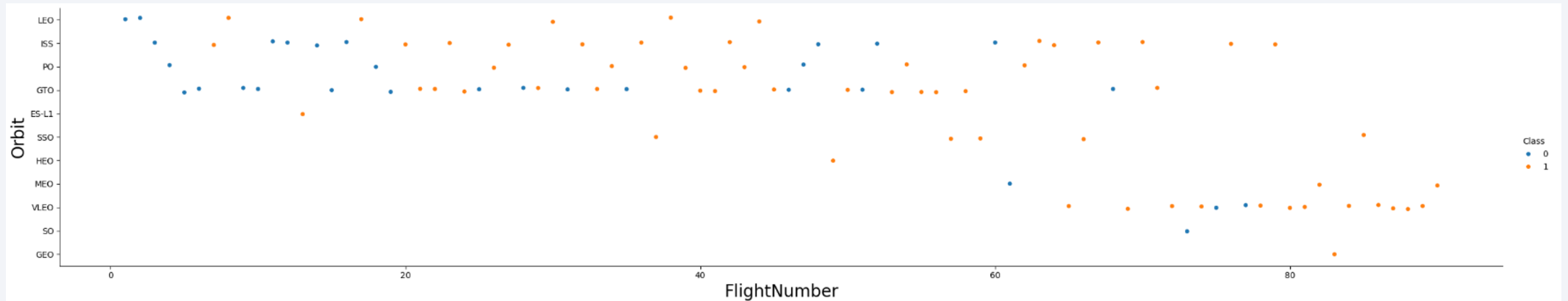
- This plot shows launch sites (CCSFS SLC 40, KSC LC 39A, and VAFB SLC 4E) with payload, and class(Failure, Success)
- VAFB SLC 4E launched little and payloadmass was little
- CCSFS SLC 40 has more launch with payloadmass between 0 to 8000

Success Rate vs. Orbit Type

- This plot shows success rate and orbit types
- ES-L1, GEO, HEO and SSO have perfect rate
- GTO has 50% and SO has 0%
- Others rate are higher than 60%

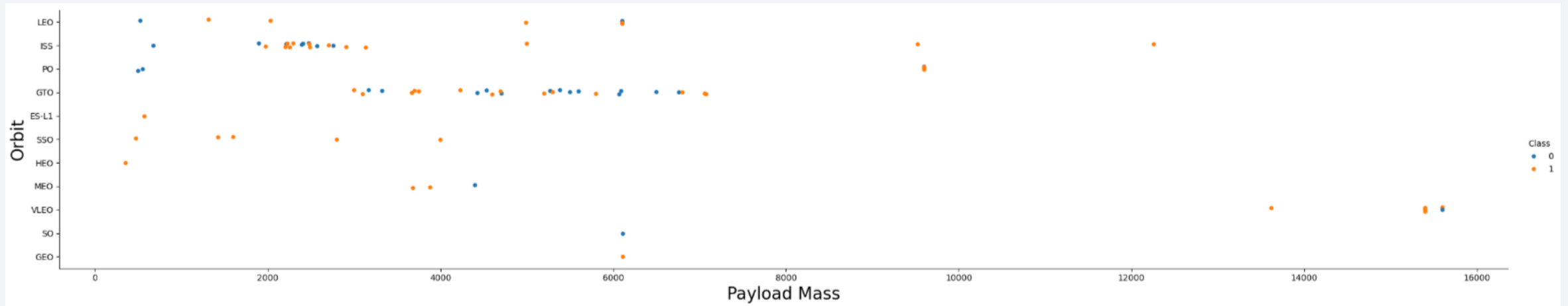


Flight Number vs. Orbit Type



- This plot shows Flight number and Orbit type, however, there is no relationship between them

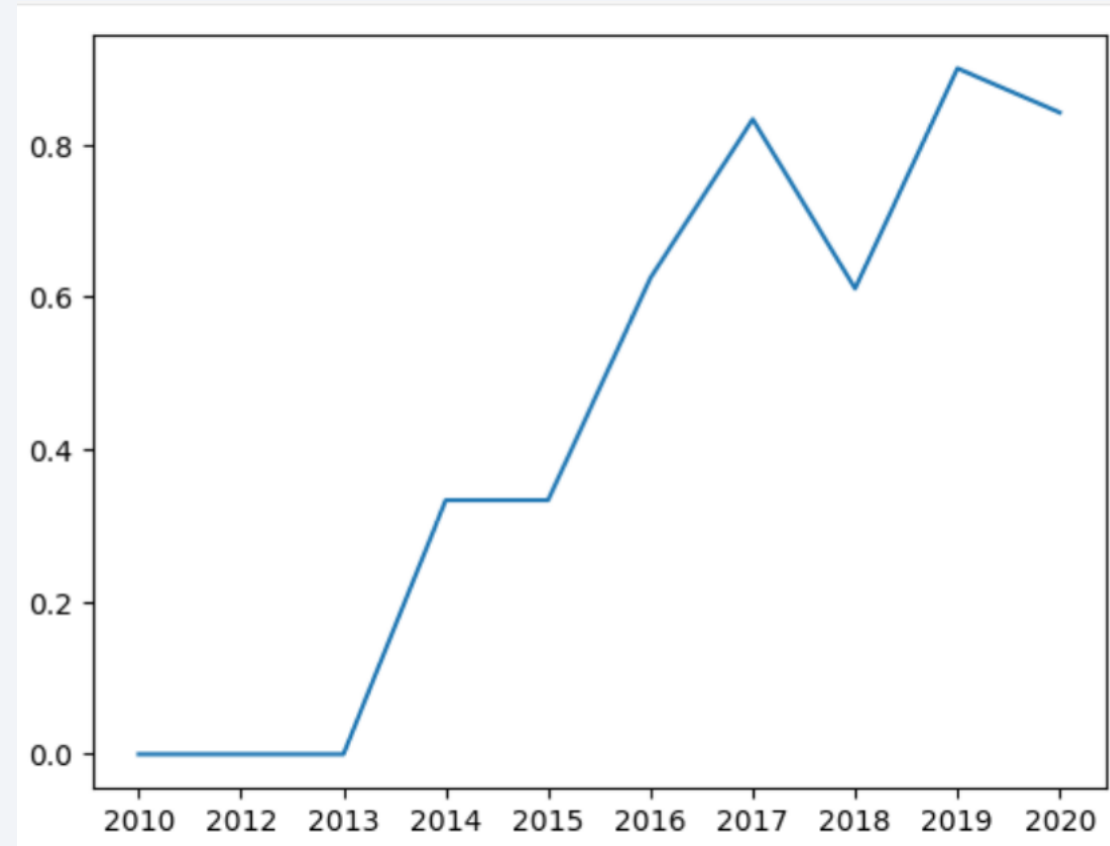
Payload vs. Orbit Type



- This plot shows payload and orbit type
- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend

- This plot shows yearly average success rate
- The success rate increasing since 2013 till 2020



All Launch Site Names

```
%sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
```

```
#SELECT DISTINCT Launch_Site
```

```
#FROM df;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
Launch_Sites
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

- DISTINCT is used to show unique launch sites

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- SELECT, LIKE, LIMIT to show launch sites begin with 'CCA' in 5 records

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Total Payload Mass(Kgs)	Customer
45596	NASA (CRS)

- the total payload carried by boosters from NASA = 45596
- SELECT, SUM were used to calculate

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1%';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Payload Mass Kgs	Customer	Booster_Version
2534.6666666666665	MDA	F9 v1.1 B1003

- the average payload mass carried by booster version F9 v1.1 = 2928.4
- SELECE, AVG were used to calculate

First Successful Ground Landing Date

```
%sql SELECT MIN(Date) FROM 'SPACEXTBL' WHERE Landing_Outcome = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
MIN(Date)
```

```
2015-12-22
```

- the dates of the first successful landing outcome on ground pad is 2015-12-22
- SELECT, MIN AVG were used to select the date

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS__KG_ > 4000 AND
```

```
* sqlite:///my_data1.db
```

Done.

Booster_Version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11 / EchoStar 105

- F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
- SELECT, DISTINCT, AND to select the name in between

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- the total number of successful mission outcomes = 100
- the total number of failure mission outcomes = 1
- SELECT, COUNT were used

Boosters Carried Maximum Payload

```
%sql SELECT "Booster_Version",Payload, "PAYLOAD_MASS_KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SF
```

* sqlite:///my_data1.db

Done.

Booster_Version	Payload	PAYLOAD_MASS_KG_
F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

- SELECT MAX were used

2015 Launch Records

```
%sql SELECT substr(Date,0,5), substr(Date,6,2), "Booster_Version", "Launch_Site", "Payload", "PAYLOAD_MASS_KG_", "Mission_Outcome", "Landing_
```

```
* sqlite:///my_data1.db
```

```
Done.
```

substr(Date,0,5)	substr(Date,6,2)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Mission_Outcome	Landing_Outcome
2015	01	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	Success	Failure (drone ship)
2015	04	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	Success	Failure (drone ship)

- SELECT, WHERE, LIKE, AND were used to filter failure landing_outcomes in drone ship for the months in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT * FROM SPACEXTBL WHERE "Landing_Outcome" LIKE 'Success%' AND (Date BETWEEN '2010-06-04' AND '2017-03-20') ORDER BY Date DESC;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-01-14	17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communications	Success	Success (drone ship)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-07-18	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2016-05-27	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom	Success	Success (drone ship)
2016-05-06	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group	Success	Success (drone ship)
2016-04-08	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)	Success	Success (drone ship)
2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	Success	Success (ground pad)

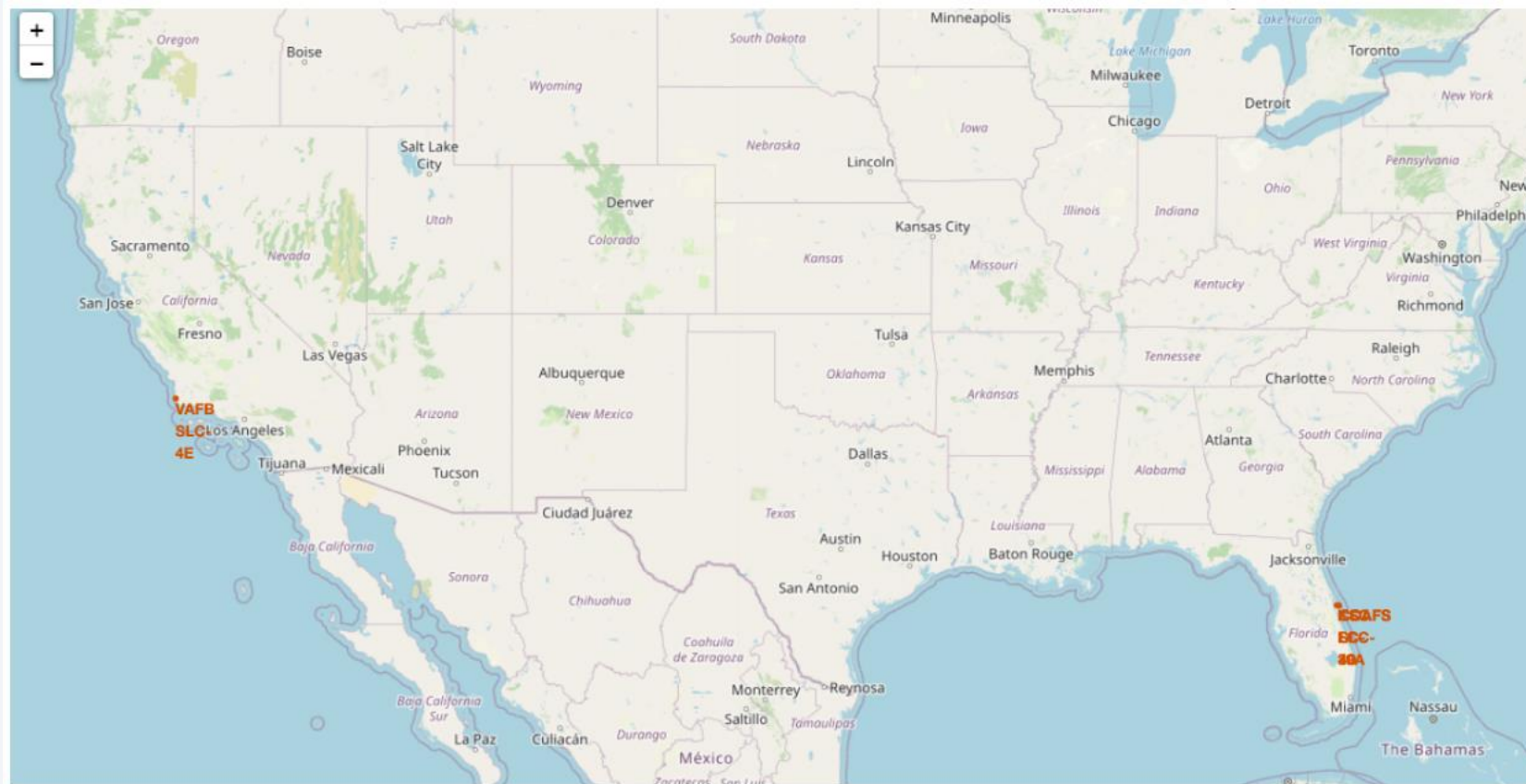
- SELECT, LIKE, AND, BETWEEN, ORDER BY were used to Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

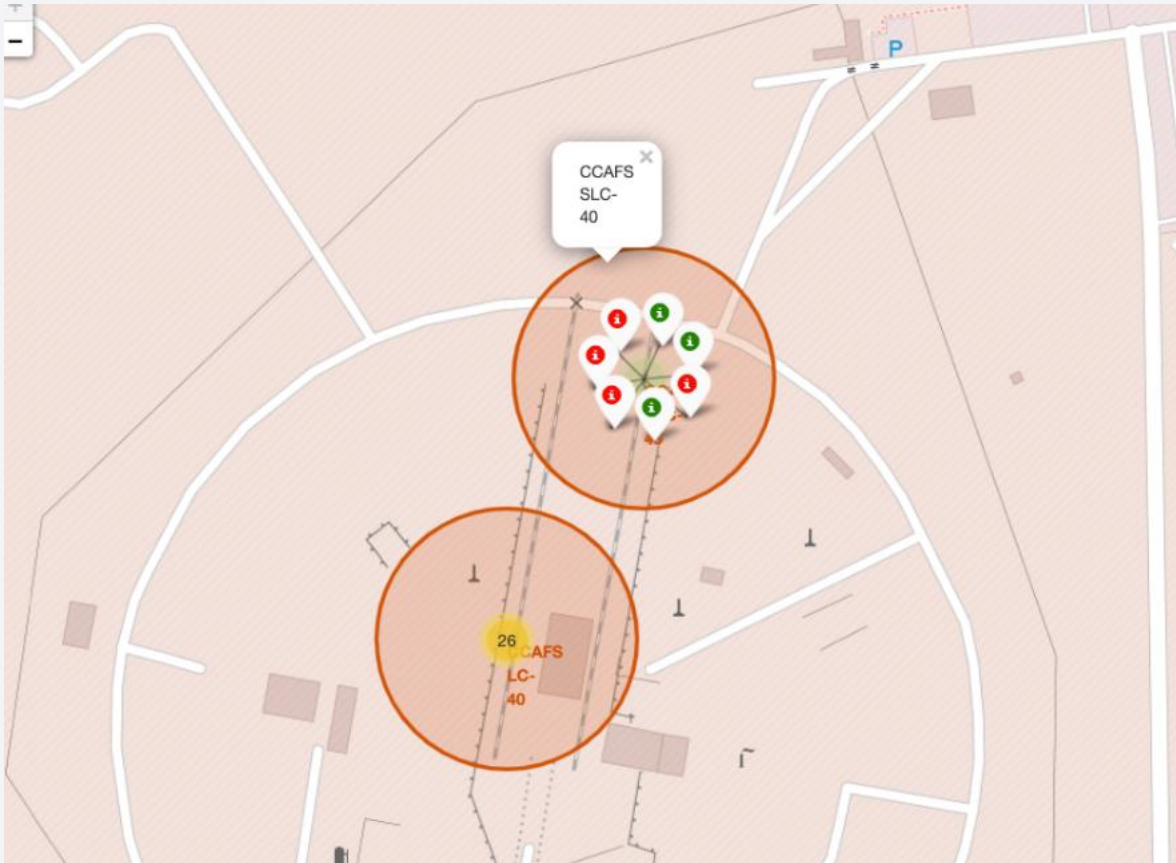
Launch Sites Proximities Analysis

All launch sites in map



- The SpaceX launch sites are in Florida and California of the USA

Colored labeled markers shows in launch sites

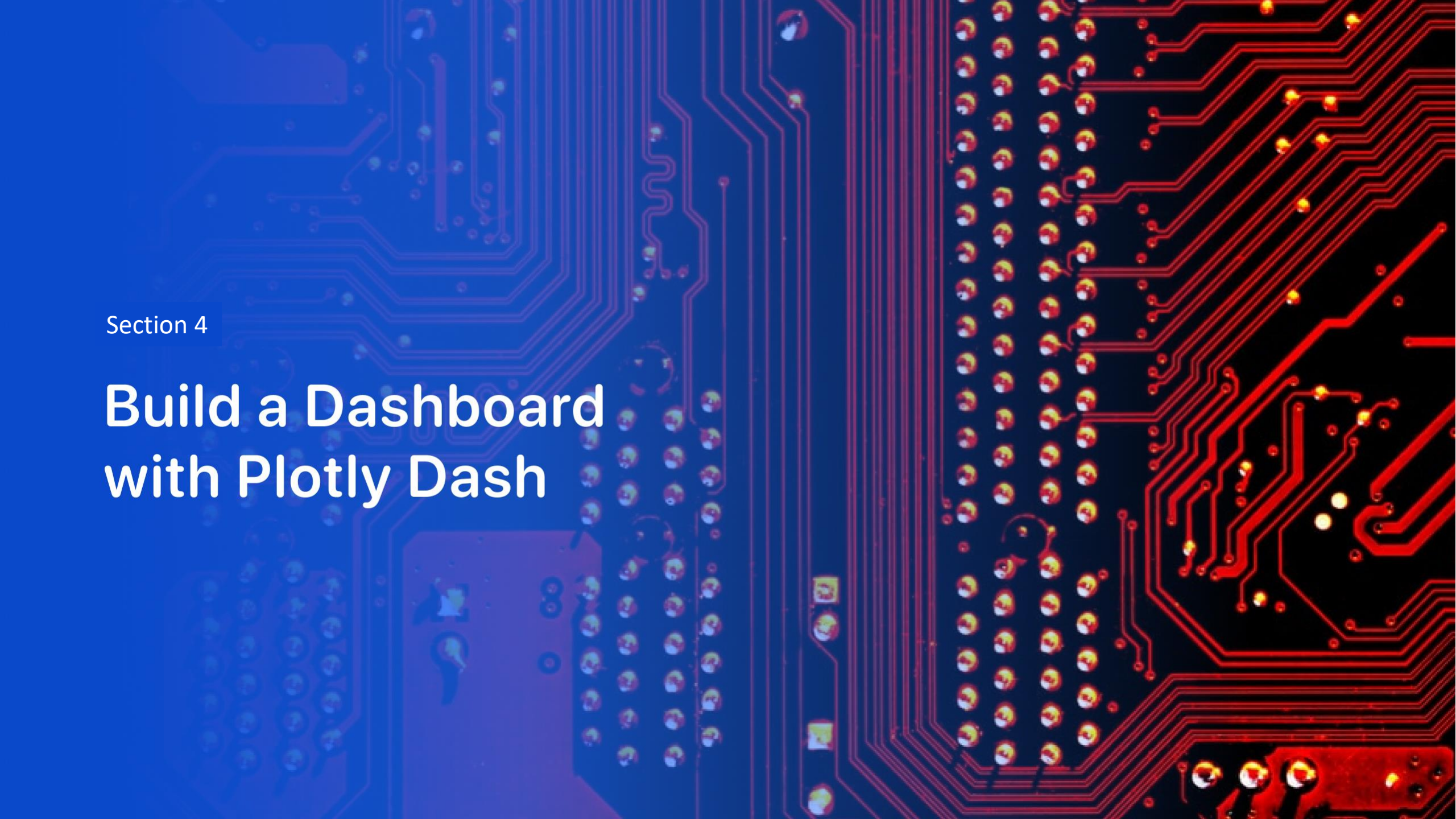


- Green marker are successful launches
- Red marker are unsuccessful launches

<Folium Map Screenshot 3>



1. Are lunch sites is close proximity to railways?
- No
2. Are lunch sites is close proximity to highway?
- No
3. Are lunch sites is close proximity to coastline?
- Yes
4. Do launch sites keep certain distance away from cities?
- Yes



Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

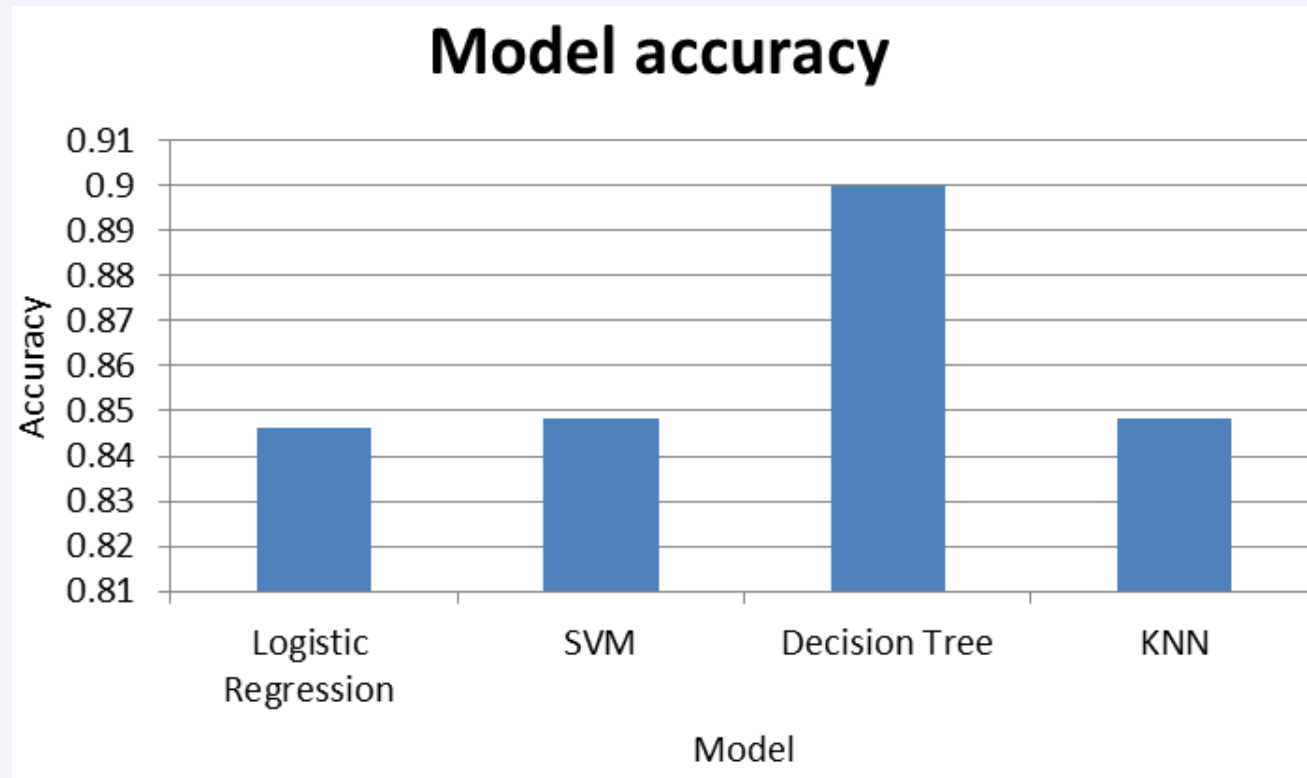
<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Section 5

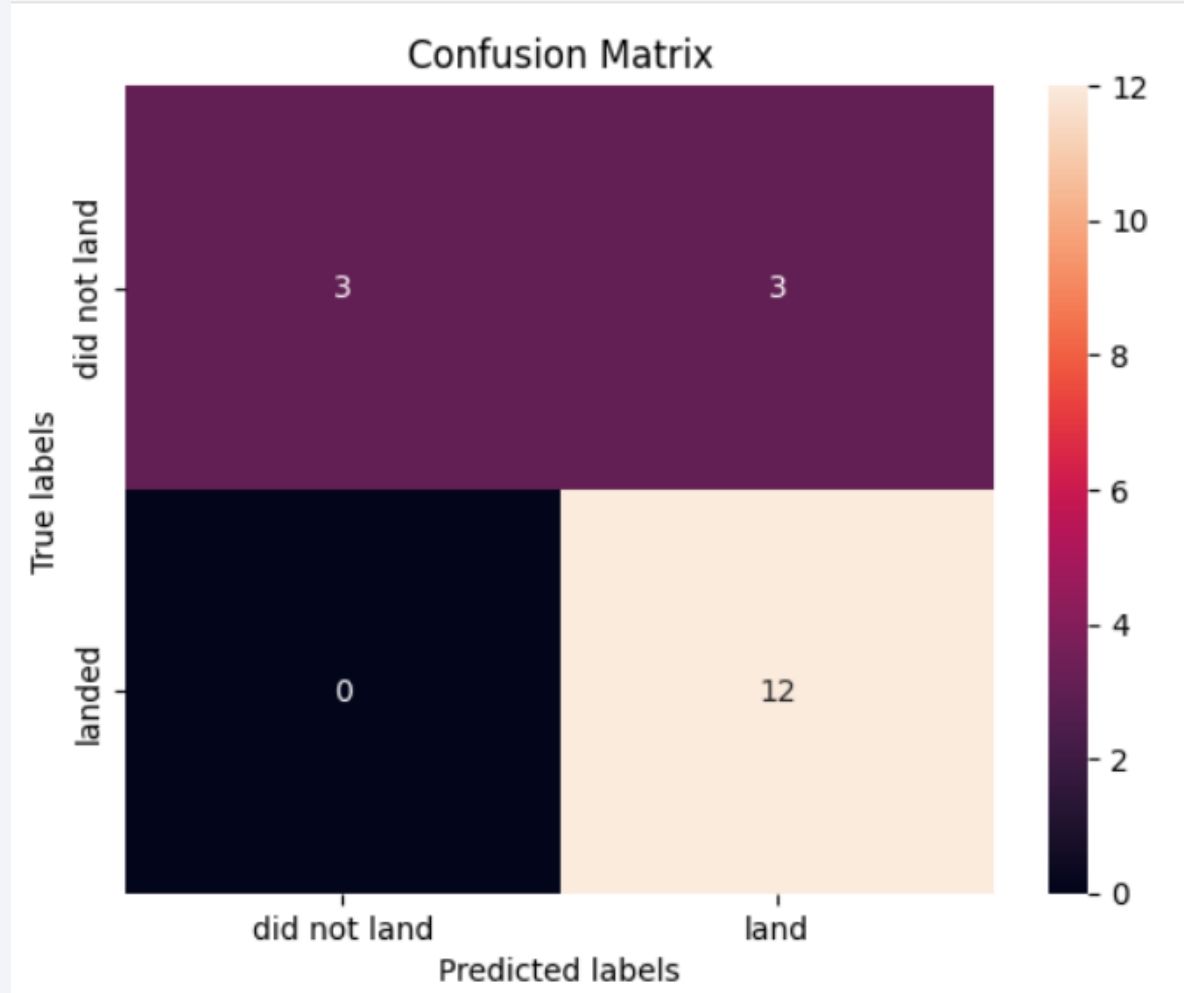
Predictive Analysis (Classification)

Classification Accuracy



- Decision tree has the better accuracy (0.9)
- Others have around 0.85 accuracy

Confusion Matrix



- This is the confusion matrix of decision tree model
- 12 landed were predicted as landed
- 3 do not landed were wrong predicted as landed

Conclusions

- The success rate of SpaceX launch mission has increased over the years
- the Orbits ES-L1, GEO, HEO, SSO, VLEO has the most success rate
- KSC LC-39A is the most successful launch site
- The decision tree model is the best machine learning algorithm in this task

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

